Date / /
Page

Fractional knapsack

```
#include <stdio.h>
#include <time.h>

struct Item
{
    float ratio;
    int value, weight;
};

void swap (struct Item a[], int i, int j)
{
    struct Item temp = a[i];
    a[i] = a[j];
    a[j] = temp;
}

void sort (struct Item a[], int n)
{
    for (int i=0; i<n-1; i++)
        for (int j=0; j<n-1-i; j++)
        {
            if (a[j].ratio < a[j+1].ratio)
            {
                swap (a, j, j+1);
            }
        }
}

float fracknap (struct Item a[], int n, int capacity)
{
    float value = 0.0;
```

```
for(int i=0; i<n; i++)
{
    if(capacity >= a[i].weight)
    {
        totalValue += a[i].value;
        capacity -= a[i].weight;
    }
    else
    {
        value += a[i].ratio * capacity;
        break;
    }
}
return value;
}

int main()
{
    int n, capacity;

    printf("Enter number of item: ");
    scanf("%d", &n);
    struct item a[n];

    for (int i=0; i<n; i++)
    {
        printf("Enter the value &
                weight of item %d: ", i+1);
        scanf("%d %d", &a[i].value, &a[i].weight);
        a[i].ratio = (float) a[i].value / a[i].weight;
    }
```

```
printf("Enter knapsack capacity: ");
scanf("%d", &capacity);
sort(a, n);
clock_t start, end; start = clock();
float maxVal = fracknap(a, n, capacity);
end = clock();

printf("Maximum profit = %.2f\n",
                    maxVal);

double time = (double)(end - start) /
                    CLOCKS_PER_SEC;

printf("Execution time: %f", time);

return 0;
}
```

22/13/4

Output:

```
Enter number of items: 3
Enter value & weight of item 1: 25 18
                             2: 24 15
                             3: 15 10
Enter knapsack capacity : 20.
Maximum profit = 31.50:

Execution time : 0.00.
```